
Algorithms – Exercise Sheet 1

Discussion on Monday, 2026-04-20

Exercise 1 (Standard Formulas)

Let $n \in \mathbb{N}$ and $q \in \mathbb{R}^+$. Prove the following statements.

(a) The Gauss Formula

$$\sum_{i=1}^n i = \sum_{i=0}^n i = \frac{n \cdot (n+1)}{2} = \Theta(n^2).$$

(b) Geometric Series

$$\sum_{i=0}^n q^i = \begin{cases} \frac{q^{n+1}-1}{q-1} & \text{for } q \neq 1 \\ n+1 & \text{for } q = 1 \end{cases} = \begin{cases} \Theta(1) & \text{for } q < 1 \\ \Theta(n) & \text{for } q = 1 \\ \Theta(q^n) & \text{for } q > 1 \end{cases} \quad \text{and} \quad \sum_{i=0}^{\infty} q^i := \lim_{n \rightarrow \infty} \sum_{i=0}^n q^i = \frac{1}{1-q} \text{ for } q < 1.$$

(The last formula also holds for $-1 < q \leq 0$.)

Moral: In big- Θ terms, the sum of a geometric series is simply the first term if the series is strictly decreasing, the last term if the series is strictly increasing, or the number of terms if the series is unchanging.

(c) Harmonic Numbers

Let $H_n := \sum_{i=1}^n \frac{1}{i}$ be the n -th harmonic number. Show that $H_n = \Theta(\ln n)$ by proving the inequalities

$$\ln n < H_n < \ln n + 1 \quad \text{for } n \geq 2.$$

Hint: $\frac{1}{i} < \int_{i-1}^i \frac{1}{t} dt < \frac{1}{i-1}$ for all $i \geq 2$.

Exercise 2 (Recursion trees)

Let $c > 0$ be a constant. Draw **recursion trees** for the following recurrences and give an asymptotic bound for each recurrence. The base case for each recurrence is $A[1] = \dots = D[1] = c$.

(a) $A[n] = 2 \cdot A[\frac{n}{4}] + c \cdot \sqrt{n}$

(b) $B[n] = 3 \cdot B[\frac{n}{3}] + c \cdot n^2$

(c) $C[n] = 5 \cdot C[\frac{n}{2}] + c \cdot n$

(d) $D[n] = D[\frac{n}{2}] + D[\frac{n}{3}] + D[\frac{n}{6}] + cn$

Useful Limit Rules

- Let $f: \mathbb{N} \rightarrow \mathbb{R}^+$ and $g: \mathbb{N} \rightarrow \mathbb{R}^+$ be functions, and let $h: \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing function such that $\lim_{n \rightarrow \infty} h(n) = \infty$.

$$\text{If } \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0, \text{ then } \lim_{n \rightarrow \infty} \frac{f(h(n))}{g(h(n))} = 0.$$

- For all $k \geq 0$ and $\ell > 1$, we have

$$\lim_{n \rightarrow \infty} \frac{n^k}{\ell^n} = 0.$$

- For all $k \geq 0$ and $\ell > 0$, we have

$$\lim_{n \rightarrow \infty} \frac{(\log n)^k}{n^\ell} = 0.$$

- L'Hôpital's rule:* Let $f: \mathbb{N} \rightarrow \mathbb{R}^+$ and $g: \mathbb{N} \rightarrow \mathbb{R}^+$ be functions such that $\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} g(n) = 0$ or $\lim_{n \rightarrow \infty} |f(n)| = \lim_{n \rightarrow \infty} |g(n)| = \infty$.

$$\text{If } \lim_{n \rightarrow \infty} \frac{\frac{d}{dn} f(n)}{\frac{d}{dn} g(n)} \text{ exists, then } \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{\frac{d}{dn} f(n)}{\frac{d}{dn} g(n)}.$$

Useful Logarithm Rules

- $\log_b (x \cdot y) = \log_b x + \log_b y$ and $\log_b (x^y) = y \cdot \log_b x$
- $x = b^{\log_b x}$, $x^{\log_b y} = y^{\log_b x}$, and $\log_y x = \frac{\log_b x}{\log_b y} = \frac{1}{\log_x y}$.
- $\frac{d}{dx} \log_b x = \frac{1}{x \cdot \ln b}$